

POKHARA UNIVERSITY
FACULTY OF SCIENCE & TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2025 Fall		
Level	B.E.	F M	100
Program	Computer	PM	45
Year/ Part	4/1	Time	3 Hrs

Subject: Network and Cyber Security

*Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.*

1. ☒ a) What is a network security model? Describe the basic components of a network security model. 7
8
☒ b) Explain the encryption and decryption process of DES. Why Double DES failed to provide adequate security.
2. ☒ a) Differentiate between Stream Cipher and Block Cipher (any 4). The ciphertext "DWWDFN" was generated using a Caesar Cipher. Find the plaintext if the key used is 3. 7
☒ b) Discuss the need for asymmetric key cryptography. Explain how public key systems overcome the key management and distribution problems of symmetric key cryptography. 8
3. ☒ a) Alice and Bob agree on a prime number $p = 23$ and generator $g = 5$. 7
 - Alice chooses private key $a = 6$
 - Bob chooses private key $b = 15$

Calculate:

 - i. Alice's public key
 - ii. Bob's public key
 - iii. The shared secret key at both ends☒ b) Explain the steps involved in RSA algorithm and discuss the basis of its security. 8
4. ☒ a) What is a MIM attack? Discuss the mechanisms used to secure communication against it. 7
8
☒ b) Define a hash function and explain its properties with examples. Also, explain the concept of collision in hash function.
5. ☒ a) Discuss common threats in web security and how they are mitigated. 7
☒ b) Explain the applications of SSL and describe the SSL/TLS handshake process. 8
6. ☒ a) Explain a scenario where broken access control and insecure session management can lead to data compromise. 7
8

b) What is data encryption at rest and in transit? Differentiate between authentication and authorization.

7. Write short notes on: (Any two)

2×5

- a) Cryptography
- b) Kerberos
- c) Pretty Good Privacy

*** Best of Luck ***